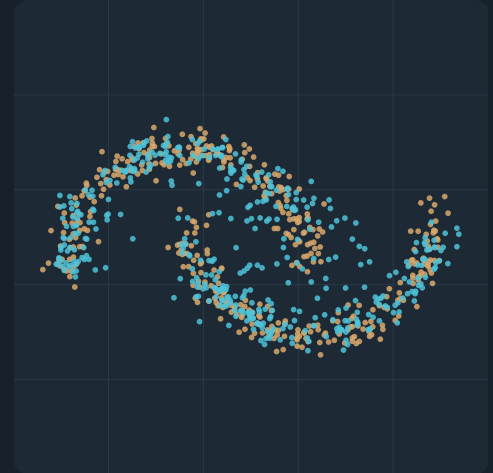


Flow Matching · declared path

The interpolation formula defines a path. **field** regresses $E[\dot{x}_t | x_t]$ — the L^2 minimizer, solving the continuity equation for this path. **flow** defines a pushforward using the field.

```
xt = t*data + (1-t)*noise
pt = prob(xt)
v = field(pt, estimator=regress(mlp(2->64->64->2)))
T = flow(v, steps=50)
plot data, (T # noise) ~ 1000
```

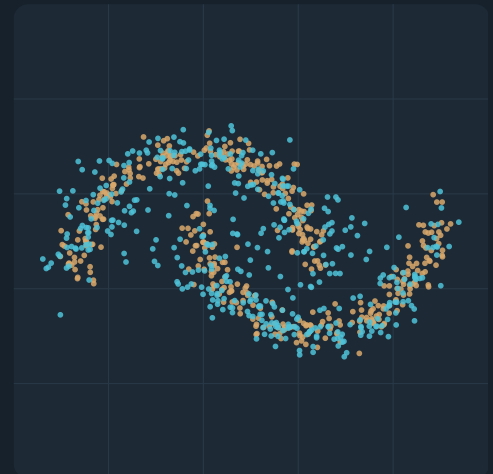


● target sample ● T # noise

Diffusion · one line away

Change the interpolation — same regression.

```
xt = sqrt(1-t*t)*noise + t*data
pt = prob(xt)
v = field(pt, estimator=regress(mlp(2->96->96->2)))
T = flow(v, steps=60)
plot data, (T # noise) ~ 800
```

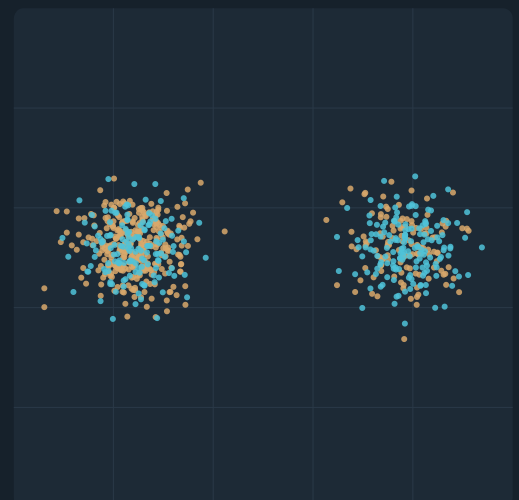


● target sample ● T # noise

SVGD · gradient flow

descent defines the steepest descent of a divergence in $P(\mathbb{R}^2)$. No regression — the field is lazily estimated each step.

```
cloud = uniform([-3,-3], [3,3]) ~ 300
target = 0.7*gaussian([-2,0], 0.4)
         | 0.3*gaussian([ 2,0], 0.4)
qt = descent(reverseKL(target), from=cloud)
v = field(qt, estimator=nw(kernel=rbf))
T = flow(v, steps=400, lr=0.8)
plot target ~ 1500, (T # cloud)
```



● target sample ● T # cloud